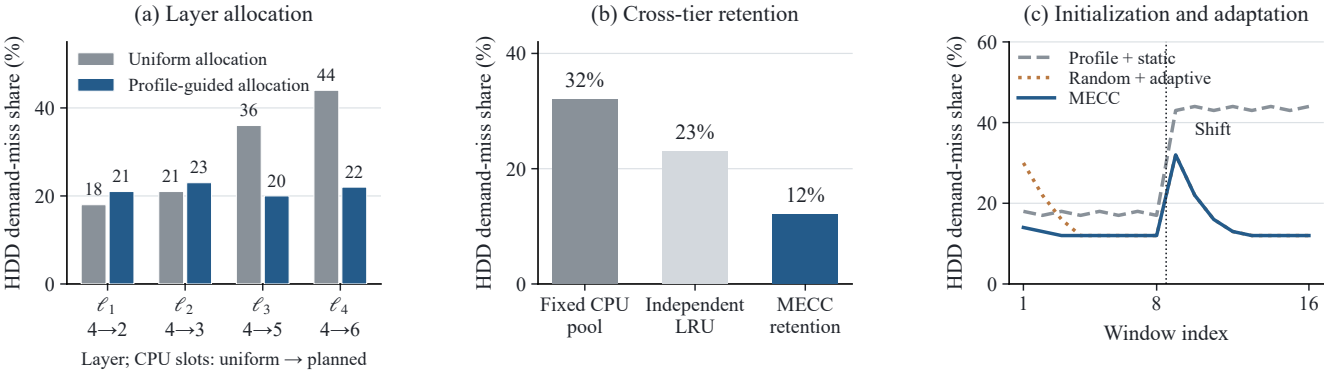
(a) Layer allocation
HDD demand-miss share (%)
Uniform allocation
Profile-guided allocation
0
20
40
18
21
21
23
36
20
44
22
ℓ_1 4→2
ℓ_2 4→3
ℓ_3 4→5
ℓ_4 4→6
Layer; CPU slots: uniform → planned
(b) Cross-tier retention
HDD demand-miss share (%)
0
20
40
32%
23%
12%
Fixed CPU pool
Independent LRU
MECC retention
(c) Initialization and adaptation
HDD demand-miss share (%)
Profile + static
Random + adaptive
MECC
Shift
0
20
40
60
1
8
16
Window index